

Date: 2026-03-06, the start of the third plague pandemic in the US in 1900.

1. The $ETS(AAdN)$ model is given by the system

$$\begin{cases} u_t \sim \mathcal{N}(0; 25) \\ b_t = 0.5b_{t-1} + 0.1u_t \\ \ell_t = \ell_{t-1} + 0.5b_{t-1} + 0.1u_t \\ y_t = \ell_{t-1} + 0.5b_{t-1} + u_t \end{cases}$$

with $\ell_{100} = 30$ and $b_{100} = 2$.

- (a) [5] Find the 95% predictive interval for y_{102} .
 - (b) [5] Find conditional probability $\mathbb{P}(y_{102} > 35 \mid \ell_{100}, b_{100})$ in terms of the standard normal distribution function.
2. The semi-annual y_t is modelled by $ETS(ANA)$ process:

$$\begin{cases} u_t \sim \mathcal{N}(0; 16) \\ s_t = s_{t-2} + 0.1u_t \\ \ell_t = \ell_{t-1} + 0.5u_t \\ y_t = \ell_{t-1} + s_{t-2} + u_t \end{cases}$$

with $\ell_0 = 20$, $s_0 = 5$ and $s_{-1} = -5$.

- (a) [5] Decompose first two observations $y_1 = 25$ and $y_2 = 20$ into level, seasonal and error components.
 - (b) [5] Is the process (y_t) stationary?
3. Consider $MA(1)$ process $y_t = 3 + u_t + 4u_{t-1}$, where u_t is a white noise process with variance 10.
- (a) [5] Find $\mathbb{E}(y_t)$, $\text{Var}(y_t)$, $\text{Cov}(y_t, y_s)$.
 - (b) [1] Is the process (y_t) stationary?
 - (c) [1] Is the process (y_t) invertible with respect to (u_t) ?
 - (d) [3] If possible represent u_t in the form $c + \sum_{i=-\infty}^{+\infty} \alpha_i y_{t+i}$.

4. Consider the equation $y_t = 0.4y_{t-1} + u_t + 0.5u_{t-1}$, where u_t is a white noise process with variance σ^2 .
- (a) [1] How many non-stationary solutions does this equation have?
 - (b) [3] How many stationary solutions of $MA(\infty)$ form with respect to (u_t) does this equation have? Find all solutions, if possible.
 - (c) [2] Rewrite all stationary solutions in $AR(\infty)$ form with respect to (u_t) , if possible.
 - (d) [4] Find $\mathbb{V}\text{ar}(y_t)$ for all stationary solutions.
5. Consider the stationary solution of the equation

$$y_t = 0.8y_{t-1} - 0.5y_{t-2} + u_t,$$

where (u_t) is a white noise.

- (a) [3] Check the existence of stationary solution.
 - (b) [4] Find the values of ACF $\rho(1), \rho(2), \rho(3)$.
 - (c) [3] Find the values of PACF $\phi(1), \phi(2), \phi(3)$.
6. For each of the processes below:
- (a) Determine whether the stationary or non-stationary process exist.
 - (b) If possible, identify its *ARIMA* order.
 - (c) If the process is non-stationary, indicate the reason and suggest a method to transform it into a stationary form.
- [2] $y_t = 0.6y_{t-1} + 0.2y_{t-2} + u_t + 0.3u_{t-1}$
 - [2] $x_t = x_{t-1} + u_t - 0.7u_{t-1}$
 - [3] $z_t = 2z_{t-1} + u_t - 2u_{t-1}$
 - [3] $w_t = 5 + 2t + 0.8w_{t-1} + u_t$

Here (u_t) is a white noise process.
